Covariance Implied Risk Factors

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Abstract

Principal component analysis extracts latent risk factors from the sample covariance matrix, whose diagonal mixes systematic and idiosyncratic variance. When idiosyncratic variances differ across assets, PCA overweights the noisiest assets, so its factors capture idiosyncratic rather than systematic risk. To correct this distortion, I apply heteroskedastic PCA, which iteratively replaces the diagonal with values implied by the covariances. This approach delivers superior out-of-sample cross-sectional pricing performance compared to standard PCA, with higher Sharpe ratios and lower pricing errors across equity and FX cross-sections. It also recovers economically interpretable factors that standard PCA leaves buried in noise. The gains are largest when the factor eigenvalues are poorly separated relative to idiosyncratic noise.

Keywords: Factor models, principal component analysis, heteroskedasticity, asset pricing, latent factors.

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1. Introduction

A fundamental insight in asset pricing theory is that expected returns should be explained by exposure to systematic risk factors. Identifying these factors has become a central pursuit in empirical finance, leading to what Harvey et al. (2015) characterize as a "factor zoo", with over 300 proposed risk factors. Recent work has sought to bring order to this chaos by summarizing pricing information with a small number of latent factors (Kozak et al., 2020; Lettau and Pelger, 2020; He et al., 2023; Bryzgalova et al., 2023). The goal is to preserve interpretability and stability while capturing the priced sources of risk.

Many of these approaches rely on principal component analysis (PCA), which is appealing because it is nonparametric, computationally tractable, and scales easily to large cross-sections and time series. However, standard PCA operates on the sample covariance matrix and implicitly assumes that idiosyncratic errors are homoskedastic. When assets exhibit heterogeneous idiosyncratic variances, as is common in equity, foreign exchange, and other markets, PCA factors become distorted. Because PCA maximizes total explained variance, it loads disproportionately on high-variance assets even when their variance reflects idiosyncratic noise rather than systematic risk (Zhang et al., 2022; Bai and Ng, 2002). Consequently, the estimated factors may reflect clusters of idiosyncratic risk instead of the underlying systematic components.

This paper addresses this problem by applying heteroskedastic PCA (heteroPCA), a recently developed method from the statistics literature (Zhang et al., 2022), to the estimation of latent risk factors in finance. HeteroPCA corrects for heteroskedasticity in idiosyncratic errors by iteratively replacing the diagonal of the sample covariance matrix with estimates implied by a low-rank approximation of the off-diagonal structure. The key insight is that systematic information resides primarily in the covariances (off-diagonal elements), while heteroskedastic noise inflates the variances (diagonal elements). By downweighting the diagonal and focusing on the systematic covariance structure, heteroPCA delivers more accurate factor estimates.

My contribution is threefold. First, I demonstrate that heteroPCA yields superior out-of-sample cross-sectional pricing performance compared to standard PCA. Across multiple equity portfolios sorted on size, book-to-market, accruals, investment, operating profitability, and other characteristics, heteroPCA delivers higher Sharpe ratios and lower root mean squared pricing errors. These improvements are economically meaningful: the method identifies factors with clearer economic content and achieves better risk-return tradeoffs.

Second, I show that the factors identified by heteroPCA exhibit stronger economic

interpretability. In cross-sections with known factor structures, such as portfolios sorted on size and accruals, heteroPCA shares the market factor with standard PCA but, unlike standard PCA, isolates a clean accrual factor rather than leaving it buried in a noisy higher-order component. By filtering out idiosyncratic noise, heteroPCA uncovers systematic risk that remains hidden under standard methods.

Third, I examine the stochastic discount factor (SDF) implied by the estimated factors. HeteroPCA consistently delivers lower Hansen-Jagannathan distances, indicating better cross-sectional pricing accuracy. The SDF-mimicking portfolios exhibit intuitive patterns: for instance, in size-accrual portfolios, heteroPCA identifies a clear quality-versus-junk tilt, shorting small high-accrual stocks and holding low-accrual positions, whereas standard PCA produces less interpretable "size barbell" patterns.

My findings highlight the practical importance of accounting for heteroskedasticity when extracting latent factors. Heteroskedastic idiosyncratic variances are pervasive across asset classes. In equity markets, firm size, industry composition, leverage, and growth opportunities all contribute to heterogeneous volatility (Fama and French, 1992; Campbell et al., 2001; Ang et al., 2006). In foreign exchange markets, interest rate differentials, sensitivity to global macro shocks, and liquidity differences generate varying levels of idiosyncratic risk across currencies (Menkhoff et al., 2012; Andersen et al., 2003; Della Corte et al., 2016). Even when assets are grouped into portfolios, as is standard practice, characteristic-driven volatility persists. My results demonstrate that correcting for this heteroskedasticity leads to more reliable and interpretable risk factors.

This paper contributes to several strands of the literature. First, it relates to the growing body of work on latent factor models and PCA-based methods in asset pricing, including instrumented PCA (Kelly et al., 2019), projected PCA (Fan et al., 2016), and risk premium PCA (Lettau and Pelger, 2020). My approach is complementary: by addressing heteroskedasticity, heteroPCA can be integrated into these existing frameworks to improve factor estimation. Second, it connects to the literature on heteroskedasticity in asset returns and its implications for estimation (Zhang et al., 2022; Bai and Ng, 2002). I show that the statistical insights from this literature have direct and substantial implications for empirical asset pricing. Third, the results relate to the literature on stochastic discount factors and Hansen-Jagannathan bounds (Hansen and Jagannathan, 1991), demonstrating that better factor estimation translates into improved SDF performance.

Finally, the approach speaks to the long-standing debate over whether expected returns are driven by firm characteristics or by covariances with systematic factors (Daniel and Titman, 1997; Kelly et al., 2019). The test assets I study are themselves characteristic-

sorted portfolios, so their cross-sectional dispersion in average returns is organized by characteristics, yet heteroPCA extracts its factors purely from the covariance structure, deliberately discarding the heteroskedastic diagonal. In this sense the method sits between the two views: it lets characteristics define the cross-section to be priced, while insisting that the priced factors be recovered from co-movement rather than from the characteristics directly. That heteroPCA sharpens, for example, an accrual dimension that standard covariance-based PCA misses suggests that much of what looks like a "characteristic" premium is in fact a systematic covariance dimension obscured by idiosyncratic-variance heterogeneity.

2. Theoretical framework and methodology

This section describes the approximate factor model underlying my analysis, explains how heteroskedasticity distorts standard PCA, and presents the heteroPCA methodology for correcting this bias.

2.1. The approximate factor model

I assume that the excess returns of N assets over T time periods follow an approximate factor model with K latent factors:

$$r_{nt} = \lambda_n^\top f_t + \varepsilon_{nt}, \qquad n = 1, \ldots, N, \;\; t = 1, \ldots, T, \tag{1}$$

where r_{nt} is the excess return on asset n at time t, $f_t \in \mathbb{R}^K$ is the vector of common factors, $\lambda_n \in \mathbb{R}^K$ is the vector of factor loadings for asset n, and ε_{nt} is the idiosyncratic error with $\mathbb{E}[\varepsilon_{nt}] = 0$ and $\mathrm{Var}(\varepsilon_{nt}) = \sigma_n^2$. In matrix form:

$$\underbrace{R}_{T\times N} = \underbrace{F}_{T\times K}\,\underbrace{\Lambda^\top}_{K\times N} + \underbrace{E}_{T\times N}, \tag{2}$$

where R contains asset returns, F the factor realizations, Λ the factor loadings, and E the idiosyncratic errors.

The approximate factor model allows for weak cross-sectional dependence in the idiosyncratic errors, but requires that the common component $\Lambda\Sigma_F\Lambda^\top$ dominates asymptotically as $N \to \infty$, where $\Sigma_F = \mathrm{Var}(f_t)$ is the factor covariance matrix. Under the assumption that factors and errors are uncorrelated ($\mathbb{E}[\varepsilon_t f_t^\top] = 0$), the covariance matrix of returns

decomposes into systematic and idiosyncratic components:

$$\Sigma = \underbrace{\Lambda \Sigma_F \Lambda^\top}_{\text{systematic}} + \underbrace{\Sigma_E}_{\text{idiosyncratic}}, \qquad \Sigma_E = \text{Var}(E_t). \tag{3}$$

The sample covariance matrix is given by

$$\widehat{\Sigma} = \frac{1}{T} \sum_{t=1}^{T} (r_t - \bar{r})(r_t - \bar{r})^\top, \tag{4}$$

where $\bar{r}$ is the time-series mean. Standard PCA extracts factors by computing the top K eigenvectors of $\widehat{\Sigma}$, which estimates the loadings Λ. The factor realizations F are then obtained by projecting returns onto these estimated loadings.

2.2. The problem of heteroskedasticity

The validity of standard PCA for factor extraction relies on the implicit assumption that idiosyncratic variances are homogeneous across assets. When this assumption holds, the largest eigenvalues of $\widehat{\Sigma}$ are driven primarily by the systematic component $\Lambda \Sigma_F \Lambda^\top$, and the top eigenvectors of $\widehat{\Sigma}$ converge to those of the true systematic covariance structure.

However, when assets exhibit heteroskedastic idiosyncratic errors, this convergence breaks down. Because PCA seeks to maximize total explained variance, it loads disproportionately on assets with high total variance, even when that variance is primarily idiosyncratic. This creates two distinct problems. First, the estimated factor loadings become distorted, with excessive weight placed on high-variance assets. Second, the factors themselves reflect a mix of systematic risk and clusters of idiosyncratic volatility, rather than capturing pure common variation.

Zhang et al. (2022) and Bai and Ng (2002) formalize this problem and show that heteroskedasticity in Σ_E introduces bias in PCA-based factor estimation. The intuition is straightforward: the diagonal elements of $\widehat{\Sigma}$ contain both systematic variance (from $\Lambda \Sigma_F \Lambda^\top$) and idiosyncratic variance (from Σ_E). When idiosyncratic variances are heterogeneous and large, they dominate the diagonal and pull the eigenvectors away from the true factor space.

A common response in applied work is to standardize returns to have unit variance before applying PCA. While this eliminates heteroskedasticity, it also distorts the covariance structure by forcing all assets to contribute equally to total variance, regardless of their true systematic importance. This approach trades one bias for another and is not guaranteed to improve factor estimation.

2.3. Heteroskedastic PCA (heteroPCA)

HeteroPCA addresses heteroskedasticity directly by recognizing that systematic information resides primarily in the off-diagonal elements of $\widehat{\Sigma}$ (the covariances), while heteroskedastic noise inflates the diagonal elements (the variances). The key insight is to replace the observed diagonal with an estimate implied by a low-rank approximation of the off-diagonal structure.

The heteroPCA algorithm proceeds as follows. Let $\Delta(A)$ denote the operator that sets the diagonal of a matrix A to zero while preserving the off-diagonals. Given the sample covariance matrix $\widehat{\Sigma}$, the number of factors K, and the number of iterations T_0, heteroPCA iteratively constructs a sequence of matrices $N^{(0)}, N^{(1)}, \ldots, N^{(T_0)}$:

1. Initialize: $N^{(0)} = \Delta(\widehat{\Sigma})$ (set diagonal to zero, keep off-diagonals)
2. For $t = 0, 1, \ldots, T_0 - 1$:
   (a) Compute the best rank-K approximation of $N^{(t)}$:
   $$\widetilde{N}^{(t)} = \sum_{k=1}^{K} \lambda_k^{(t)} u_k^{(t)} u_k^{(t)\top},$$
   where $\lambda_k^{(t)}$ and $u_k^{(t)}$ are the k-th eigenvalue and eigenvector of $N^{(t)}$.
   (b) Update the matrix by replacing the diagonal:
   $$N_{ij}^{(t+1)} = \begin{cases} \widetilde{N}_{ii}^{(t)}, & i = j, \\ \widehat{\Sigma}_{ij}, & i \neq j. \end{cases}$$
3. Output: The top K eigenvectors of $N^{(T_0)}$.

Algorithm 1: HeteroPCA

The algorithm preserves the off-diagonal covariances from the data while iteratively replacing the diagonal with values consistent with a low-rank factor structure. This downweights idiosyncratic variance and focuses estimation on the systematic covariance structure. Zhang et al. (2022) prove that under heteroskedasticity, heteroPCA reduces bias in eigenvector estimation and delivers more accurate recovery of the factor space compared to standard PCA.

The intuition behind heteroPCA is that the off-diagonals contain information about how assets co-move due to common factors, while the diagonals are contaminated by

heteroskedastic noise. By enforcing consistency between the diagonal and a rank-K approximation of the off-diagonal structure, heteroPCA effectively estimates what the diagonal should be if the data were generated purely by a low-rank factor model. This approach avoids the pitfalls of standardization (which distorts covariances) while directly addressing the source of the bias.

3. Data and empirical implementation

3.1. Data

The empirical analysis uses a broad set of cross-sections, spanning U.S. equity portfolios and currency portfolios, to assess the performance of heteroPCA relative to standard PCA. Studying several asset classes guards against the possibility that any improvement is specific to a particular sorting scheme or market. I consider six datasets.

First, I use double- and triple-sorted equity portfolios from Kenneth French's Data Library. These portfolios sort stocks based on pairs or triples of firm characteristics including size, book-to-market (BM), accruals (ACC), investment (INV), operating profitability (OP), short-term reversal (STREV), momentum (MOM), idiosyncratic volatility (IVOL), and total volatility (VOL). The double-sorted portfolios contain $N = 25$ portfolios, while the triple-sorted portfolios contain $N = 32$ portfolios. The sample period runs from November 1963 to June 2025 at monthly frequency.

Second, I use the Asset Pricing Tree (AP-Tree) portfolios from Bryzgalova et al. (2025). These portfolios are constructed from ten firm characteristics using the AP-Tree algorithm, which generates tree-based partitions designed to capture cross-sectional variation in expected returns. I use both the Tree10 and Tree40 versions, which contain $N = 60$ portfolios each but reflect different pruning strategies. The sample period runs from January 1964 to December 2016.

Third, I reconstruct the extreme anomaly portfolios in the spirit of Kozak et al. (2020). For each of a large set of anomaly characteristics I form value-weighted decile portfolios from CRSP and retain the extreme (top and bottom) deciles, which carry the bulk of the cross-sectional spread in average returns ($N = 76$). Retaining only the extreme deciles is deliberate: aggregating to coarser groupings compresses the cross-sectional dispersion in mean returns and mechanically shrinks the differences between methods, so the extreme portfolios are where the comparison is informative.

Fourth, I use the theme-aggregated portfolios from the global factor dataset of Jensen et al. (2023). The dataset covers many countries. I take capped value-weighted portfolios for the US equity market, grouping them into 13 themes: accruals, debt issuance, low

leverage, low risk, investment, momentum, profitability, profit growth, seasonality, quality, size, short-term reversal and value, each split into terciles ($N = 39$). These portfolios summarize a very large characteristic set into a compact and highly correlated cross-section.

Fifth, I use currency portfolios for the G10 currencies, sorted on nine characteristics: carry, value, three- and twelve-month momentum and their shorter-horizon variants, volatility, skewness, and kurtosis. These are obtained from Della Corte and Riddiough (2025). For each characteristic I retain the extreme (first and fifth) quintile baskets, giving $N = 18$ portfolios over November 1983 to March 2026, the same sample of countries and start date used by Lustig et al. (2011). Currency portfolio returns are excess returns by construction, as each basket is a self-financing position funded at the domestic short rate.

Sixth, I construct a balanced panel of $N = 233$ individual stocks from CRSP. The sample period runs from May 1972 to December 2024. It is important to emphasize that this balanced panel reflects strong survivorship bias, as it includes only firms that survived the entire sample period. These firms are therefore likely to be large, mature, and relatively homogeneous, which limits the scope for heteroskedasticity corrections. I include it as a robustness check rather than as a primary cross-section.

For all datasets, I compute monthly excess returns over the risk-free rate. Following standard practice in the factor model literature, I demean returns before applying PCA or heteroPCA.

3.2. Implementation of PCA and heteroPCA

I implement both standard PCA and heteroPCA to extract $K = 3$ latent factors from each cross-section. For PCA, I compute the sample covariance matrix $\widehat{\Sigma}$ and extract the top three eigenvectors, which serve as estimates of the factor loadings $\widehat{\Lambda}$. The factor realizations $\widehat{F}$ are obtained by projecting returns onto these loadings.

For heteroPCA, I follow the algorithm described in Subsection 2.3, setting the number of iterations to $T_0 = 5$. This choice balances computational efficiency with convergence of the iterative procedure. As with PCA, the estimated loadings $\widehat{\Lambda}$ are given by the top three eigenvectors of the final matrix $N^{(T_0)}$, and the factor realizations are obtained by projection.

To ensure comparability across methods, I normalize the signs of the factors so that each has a positive mean. This normalization is standard in the factor model literature and facilitates interpretation.

3.3. Out-of-sample evaluation framework

To assess the performance of PCA and heteroPCA, I conduct an out-of-sample evaluation using rolling estimation windows, following the procedure of Lettau and Pelger (2020).

I use a rolling window of $T = 240$ months (20 years) to estimate factors and loadings. For each window ending at time t, I estimate $\widehat{F}_t$ and $\widehat{\Lambda}_t$ using data from $t - 239$ to t. I then run time-series regressions for each asset n:

$$r_{t,n} = \alpha_n + \widehat{F}_t B_n^\top + e_{t,n}, \tag{5}$$

yielding estimated loadings $\widehat{B} = [B_1, \ldots, B_N]^\top$ and pricing errors $\widehat{\alpha} = [\alpha_1, \ldots, \alpha_N]^\top$.

Next, I construct the projection matrix onto the factor span:

$$P_t = B_t (B_t^\top B_t)^{-1} B_t^\top. \tag{6}$$

I use this projection to decompose the realized next-period returns R_{t+1} into a component explained by the factors and an out-of-sample pricing error:

$$\widehat{R}_{t+1|t} = R_{t+1} P_t, \qquad \widehat{\alpha}_{t+1} = R_{t+1}(I - P_t). \tag{7}$$

The out-of-sample pricing error $\widehat{\alpha}_{t+1}$ measures the portion of realized returns that the model fails to explain when loadings are treated as constant one step ahead.

I summarize out-of-sample performance using three metrics. First, I compute the root mean squared pricing error:

$$\mathrm{RMS}_\alpha = \sqrt{\frac{1}{N} \sum_{n=1}^{N} \bar{\alpha}_n^2}, \tag{8}$$

where $\bar{\alpha}_n$ is the time-series average of the out-of-sample pricing errors for asset n. This metric captures the average magnitude of mispricing across assets.

Second, I compute the average share of unexplained variance:

$$\bar{\sigma}_e^2 = \frac{1}{N} \sum_{n=1}^{N} \frac{\mathrm{Var}(e_{t,n})}{\mathrm{Var}(r_{t,n})}, \tag{9}$$

which measures the fraction of total return variance not explained by the factors. Lower values indicate better time-series fit.

Third, I evaluate the out-of-sample Sharpe ratio of the tangency portfolio implied by the factors. In each window, I estimate the mean and covariance of the factors and

compute the Markowitz tangency portfolio in factor space. I then map this portfolio to asset space using the factor-mimicking loadings $\widehat{\Lambda}_t$ and evaluate the realized return at $t+1$. The Sharpe ratio is computed from the full out-of-sample time series of realized portfolio returns.

3.4. Stochastic discount factor diagnostics

To assess the economic content of the estimated factors, I examine the implied stochastic discount factor (SDF). For factors $\widehat{F}$ and loadings $\widehat{\Lambda}$ estimated over the full sample, I compute the vector of risk prices:

$$\widehat{\lambda} = \widehat{\Sigma}_F^{-1} \widehat{\mu}_F, \tag{10}$$

where $\widehat{\mu}_F$ and $\widehat{\Sigma}_F$ are the sample mean and covariance of the factors. These risk prices correspond to the weights of the mean-variance tangency portfolio in the factor space.

The SDF time series is then constructed as:

$$\widetilde{m}_t = \widehat{F}_t^\top \widehat{\lambda}. \tag{11}$$

I also compute the SDF-mimicking portfolio in asset space:

$$\widehat{w}_{\text{SDF}} = \widehat{\Lambda}(\widehat{\Lambda}^\top \widehat{\Lambda})^{-1} \widehat{\lambda}. \tag{12}$$

This portfolio represents the asset-space implementation of the SDF and provides economic intuition about which assets the model views as risky.

To quantify the pricing accuracy of the SDF, I compute the Hansen-Jagannathan distance (Hansen and Jagannathan, 1991):

$$d_{\text{HJ}} = \sqrt{\widehat{\alpha}^\top \widehat{\Sigma}_e^{-1} \widehat{\alpha}}, \tag{13}$$

where $\widehat{\alpha}$ is the vector of in-sample pricing errors from the time-series regressions and $\widehat{\Sigma}_e = \widehat{\text{Cov}}(e_t)$ is the covariance matrix of the residuals. The HJ distance measures how far the SDF is from correctly pricing all assets: smaller values indicate better cross-sectional fit.

4. Empirical results

4.1. Out-of-sample pricing performance

Table 1 presents the out-of-sample performance of PCA and heteroPCA across all cross-sections. The results reveal a consistent pattern: heteroPCA delivers superior cross-

sectional pricing performance compared to standard PCA, as measured by higher Sharpe ratios and lower root mean squared pricing errors (RMS_α). However, heteroPCA exhibits slightly worse time-series fit, reflected in higher average unexplained variance ($\bar{\sigma}_e^2$).

[Table 1 about here.]

For the AP-Tree portfolios, heteroPCA achieves out-of-sample Sharpe ratios of 0.52 (Tree10) and 0.57 (Tree40), substantially higher than the 0.18 and 0.26 delivered by standard PCA. Similarly, heteroPCA reduces RMS_α to 0.68 and 0.76, compared to 0.85 and 0.90 for PCA. These improvements are economically meaningful and indicate that heteroPCA identifies factors with stronger cross-sectional pricing power.

Across the double-sorted portfolios, heteroPCA outperforms PCA in the majority of cases. For instance, in the Size & Book-to-Market portfolios, heteroPCA delivers a Sharpe ratio of 0.28 versus 0.15 for PCA, while reducing RMS_α from 0.16 to 0.12. Similar patterns emerge for Size & Accruals (Sharpe ratio 0.22 vs 0.13), Size & Investment (0.31 vs 0.20), and Size & Idiosyncratic Volatility (0.29 vs 0.21). The improvements are robust across most characteristic sorts, with exceptions primarily in portfolios where both methods perform similarly.

On the reconstructed KNS extreme anomaly portfolios, heteroPCA raises the out-of-sample Sharpe ratio from 0.12 to 0.18 and slightly lowers RMS_α. On the theme-aggregated JKP factor portfolios, the gain is large, with the Sharpe ratio rising from 0.10 to 0.21. Most notably, the result carries over to a different asset class entirely: for the G10 currency portfolios, heteroPCA delivers a Sharpe ratio of 0.12 against just 0.04 for standard PCA. While the absolute Sharpe ratios in currencies are modest, the relative improvement is substantial and demonstrates that the benefit of correcting for heteroskedasticity is not an artifact of U.S. equity data or of any particular sorting scheme. In total, heteroPCA improves the out-of-sample Sharpe ratio in sixteen of the eighteen cross-sections studied.

The time-series fit, measured by $\bar{\sigma}_e^2$, is consistently better for standard PCA. This is expected because PCA maximizes total explained variance, whereas heteroPCA maximizes a low-rank approximation of the covariance structure after filtering out heteroskedastic diagonal elements. Thus, PCA naturally explains more of the time-series variation, but at the cost of loading on idiosyncratic noise. HeteroPCA trades off some time-series fit to focus on the systematic covariance structure, resulting in better cross-sectional pricing.

For individual stocks, heteroPCA does not outperform PCA. The Sharpe ratios are similar (0.14 vs 0.15), and RMS_α is only marginally lower (0.38 vs 0.39). This result is consistent with the strong survivorship bias in the balanced panel of stocks: firms

that survived from 1972 to 2024 are likely large, mature, and relatively homogeneous in their idiosyncratic volatility. Consequently, there is limited scope for heteroskedasticity corrections to improve factor estimation.

Taken together, these results demonstrate that accounting for heteroskedasticity delivers substantial gains in cross-sectional pricing performance. HeteroPCA identifies factors with stronger economic content and achieves better out-of-sample Sharpe ratios, albeit with a modest reduction in time-series fit, a tradeoff that appears economically favorable given the improvements in risk-adjusted returns.

4.2. What heteroPCA changes: subspace rotation and loading stability

Before turning to economic interpretation, I ask what the diagonal correction actually does to the estimated factors, and whether the gains come at the cost of unstable loadings. I focus on the Size & Accruals cross-section, but the patterns are representative.

The improvement is not a mere relabeling of the same factors, and it does not come at the cost of erratic loadings. I make both points with a single object, the rotation of the estimated factor subspace measured by principal angles.[2] I summarize the K estimated factors by the subspace their loading vectors span rather than by the individual loadings, because when two eigenvalues are close the individual factors are only weakly identified while the subspace they span remains well defined. Comparing the two methods within each window, Figure 1 plots the three principal angles between their factor subspaces over the rolling windows. The two methods share the market direction almost exactly, with the smallest principal angle staying near zero throughout, and differ on the two weaker directions, where the angles sit at medians of about twenty-five and eighty degrees. HeteroPCA keeps the dominant source of comovement, which is well identified regardless of heteroskedasticity, and reorients the higher-order factors away from the idiosyncratic-noise directions where the diagonal contamination matters most.

[Figure 1 about here.]

Comparing each method to itself one window earlier turns the same measure into

[2]Let $\widehat{\Lambda}_a$ and $\widehat{\Lambda}_b$ be two $N \times K$ loading matrices, either from the two methods in the same window or from one method in consecutive windows, and let Q_a and Q_b be orthonormal bases for their column spaces, obtained for example by a QR decomposition. The cosines of the K principal angles between the two subspaces are the singular values $\sigma_1 \geq \cdots \geq \sigma_K$ of the $K \times K$ matrix $Q_a^\top Q_b$, and the angles are $\theta_i = \arccos(\sigma_i)$, ordered from smallest to largest within $[0°, 90°]$. A smallest angle near zero means the two subspaces share a common direction, while a largest angle near ninety degrees means one subspace contains a direction almost orthogonal to the other. I report the largest principal angle, $\arccos(\sigma_K)$, in degrees. Because it depends only on the column spans, this measure is invariant to the order, sign, and within-subspace rotation of the factors, so it cannot be moved by an eigenvalue crossing that merely swaps the labels of two factors.

a stability diagnostic. Figure 2 plots the window-to-window rotation of each method's factor subspace. In normal times heteroPCA is markedly the more stable of the two. Its subspace barely moves, with a median rotation of about 0.2 degrees per window against 0.9 for PCA, which drifts more or less continuously and most visibly through the 1990s. Replacing the diagonal does not make the loadings erratic, and if anything it anchors them. HeteroPCA's instability is instead concentrated in a few discrete episodes, at the dot-com turn around 2000, around 2003, and around 2021, where the subspace rotates by up to ninety degrees in a single step as the cross-sectional variance environment shifts abruptly. Because the measure compares subspaces rather than individually labeled factors, these spikes are genuine reconfigurations of the factor structure, not the relabeling artifacts that a factor-by-factor comparison would show when two eigenvalues cross.

[Figure 2 about here.]

Figure 3 makes the same point at the level of individual portfolios, in the style of a time-stability plot. For each factor it overlays the loading vector estimated in every rolling window, with the portfolios on the horizontal axis, comparing heteroPCA in the top row with PCA in the bottom row. Each window's loadings are first rotated onto the whole-sample loadings, so that tightly overlapping lines indicate loadings that are stable through time. The first factor loadings are nearly identical across windows under both methods, while the second and third factor loadings show wider spread only in the structural-break episodes, with the PCA loadings displaying substantially greater dispersion than their heteroPCA counterparts.

[Figure 3 about here.]

The mechanism behind the rotation is the diagonal correction itself, which has both a cross-sectional and a time-series dimension. Across portfolios, Figure 4 plots each portfolio's original diagonal, its total variance, against the covariance-implied diagonal that heteroPCA substitutes in its place. The reduction is most pronounced for the small, extreme-accrual portfolios, whose variance is largely idiosyncratic. On the Size & Accruals portfolios the diagonal is shrunk by about eighteen percent on average and by as much as ninety-six percent for these most extreme portfolios. Over time, Figure 5 shows that the correction is present in every window, with the average shrinkage moving between about twelve and twenty percent, and that it deepens around the structural breaks where the cross-sectional variance environment shifts. It is these heavily corrected portfolios that subsequently receive the large loadings on the rotated factors, the channel through which the correction produces the more interpretable factors documented below.

[Figure 4 about here.]

[Figure 5 about here.]

A useful statistic for understanding when heteroPCA is most beneficial is the noise-to-gap ratio, defined as the average covariance diagonal divided by the eigengap between the (K)-th and $(K+1)$-th eigenvalues. This quantity appears directly in the Davis–Kahan bound on eigenvector estimation error and therefore measures the difficulty of recovering the factor space. When the ratio is large, the factor eigenvalues are only weakly separated relative to the magnitude of idiosyncratic noise, making the leading eigenvectors poorly identified. In such cases, standard PCA is more susceptible to distortions from noise-inflated diagonal entries. When the ratio is small, the factors are well separated and PCA already recovers the factor space accurately, leaving little scope for improvement from diagonal correction. HeteroPCA should therefore provide its largest gains in high-ratio windows.[3]

Figure 6 illustrates this relationship for Size & Accruals. Splitting the rolling windows into terciles of the noise-to-gap ratio, the out-of-sample Sharpe ratio of the heteroPCA factor portfolio rises far faster than that of PCA as the ratio increases. The heteroPCA advantage grows from about 0.06 in the low tercile to 0.78 in the high tercile, roughly an order of magnitude. The diagonal correction is most valuable exactly when the eigenvectors are most fragile.

[Figure 6 about here.]

Table 2 repeats this split across the cross-sections. The median heteroPCA Sharpe gain rises monotonically from 0.04 in the low tercile to 0.26 in the middle tercile and 0.55 in the high tercile, and the gain exceeds its low-tercile value in eleven of the sixteen panels. The pattern is clearest in the cross-sections whose noise-to-gap ratio reaches a genuinely high level, such as the size-and-accruals, profitability, and tree panels. It is weak or absent in panels that never leave the low-ratio regime. The currency cross-section is the clearest such case. Its noise-to-gap ratio never exceeds about three, so it remains in the region where both methods identify the factors comparably and the within-panel ordering carries little signal. This follows naturally from the mechanism. A cross-section that never faces a hard eigengap has little to gain from the correction, which is also why the absolute heteroPCA advantage on currencies is modest.

[3] A simpler signal-to-noise ratio based on the average diagonal and the (K)-th eigenvalue alone ignores the eigengap and therefore does not capture this identification problem. In the data, it also exhibits little time variation.

[Table 2 about here.]

4.3. Factor interpretability

Beyond statistical performance, a key question is whether heteroPCA delivers factors with clearer economic interpretation. To examine this, I again focus on the Size & Accruals portfolios, which have a well-understood factor structure corresponding to market, size, and accrual risk.

Figure 7 presents the factor loadings estimated by PCA and heteroPCA. For both methods, the first factor exhibits similar loadings across all size-accrual portfolios, consistent with a market factor. The second PCA factor captures a size effect, with positive loadings on small portfolios and negative loadings on large portfolios. The second heteroPCA factor instead loads overwhelmingly on the two small, extreme-accrual baskets, with smaller opposite-signed loadings on the intermediate small-cap baskets and small loadings on the larger portfolios, isolating the common variation of the most idiosyncratic portfolios rather than a smooth small-minus-big gradient.

[Figure 7 about here.]

The key difference emerges in the third factor. For standard PCA, the third factor exhibits no clear pattern across the size-accrual space. In contrast, heteroPCA's third factor loads positively on low-accrual portfolios and negatively on high-accrual portfolios, cleanly identifying an accrual factor analogous to the Fama-French accrual anomaly. This indicates that heteroPCA successfully isolates the accrual dimension that standard PCA fails to detect.

The improved interpretability of heteroPCA stems directly from its treatment of heteroskedasticity. Small, extreme-accrual portfolios exhibit high idiosyncratic volatility, which causes standard PCA to avoid loading heavily on them. By downweighting this idiosyncratic variance, heteroPCA reveals the systematic covariance structure underlying these portfolios, allowing the third factor to capture the accrual premium.

These results demonstrate that heteroPCA is not merely loading on noise, it uncovers latent systematic dimensions that remain hidden under standard PCA. Yet, the same portfolios with reduced variance receive large factor loadings after correction, because once their excess variance is stripped away, their shared co-movement becomes visible. Standard PCA misses this systematic dimension. By focusing on the off-diagonal covariance structure, heteroPCA identifies factors with clearer economic meaning and stronger alignment with known risk premia.

4.4. Stochastic discount factor

I now examine the stochastic discount factor (SDF) implied by the estimated factors. Table 3 reports the Hansen-Jagannathan distance for PCA and heteroPCA across all portfolios. HeteroPCA delivers a lower (or equal) d_{HJ} in the large majority of panels, indicating better cross-sectional pricing accuracy.

[Table 3 about here.]

The improvement is particularly pronounced in portfolios sorted on Size & Book-to-Market (0.27 vs 0.35), Size & Accruals (0.26 vs 0.31), Size & Momentum (0.29 vs 0.35), and Size & Idiosyncratic Volatility (0.34 vs 0.38). For these cross-sections, heteroPCA reduces the HJ distance by 15–30%, suggesting that the factors more accurately capture the sources of systematic risk. For the larger anomaly cross sections, KNS and JKP, heteroPCA essentially matches PCA. For FX portfolios, there is a small relative gain in heteroPCA (0.22 vs 0.24).

The economic interpretation of the SDF becomes clearer when I examine the SDF-mimicking portfolio weights. Figure 8 presents these weights for the Size & Accruals portfolios under both PCA and heteroPCA.

[Figure 8 about here.]

Under standard PCA, the SDF exhibits a "size barbell" pattern: it is long in both big and small stocks (with the strongest positions in low-accrual portfolios) and short in mid-cap stocks. This pattern lacks clear economic interpretation and suggests that PCA is capturing a mix of systematic and idiosyncratic risk.

In contrast, heteroPCA delivers a much cleaner pattern. The SDF is concentrated short in small, high-accrual portfolios and maintains small long positions elsewhere, particularly in low-accrual stocks. Big stocks are nearly neutral. This represents a clear "quality versus junk" trade: shorting small stocks with extreme accruals (which tend to have poor future performance) and holding positions in higher-quality stocks with moderate accruals.

The difference between the two approaches reflects their treatment of idiosyncratic variance. Standard PCA loads on total variance, leading to complex portfolio weights that reflect both systematic risk and idiosyncratic volatility clusters. HeteroPCA replaces the noisy diagonal with a rank-K imputation from the off-diagonal covariance structure, reorienting the SDF toward the systematic low-versus-high accrual dimension. This yields a more intuitive economic interpretation, a premium for bearing accrual risk concentrated in small-cap stocks, and delivers lower pricing errors.

These results reinforce the main findings from the out-of-sample analysis: by filtering out heteroskedastic noise and focusing on the systematic covariance structure, heteroPCA identifies factors and SDFs that better reflect the underlying sources of risk in the cross-section of asset returns.

5. Conclusion

This paper demonstrates that accounting for heteroskedasticity in idiosyncratic errors substantially improves the estimation of latent risk factors in asset pricing applications. Building on recent developments in the statistics literature, I apply heteroskedastic PCA (heteroPCA) to correct for the distortions that arise when standard PCA is applied to returns with heterogeneous idiosyncratic variances. This approach iteratively replaces the diagonal of the sample covariance matrix with estimates implied by the off-diagonal structure, downweighting idiosyncratic noise while preserving the systematic covariance information.

The empirical results demonstrate three main findings. First, heteroPCA delivers superior out-of-sample cross-sectional pricing performance compared to standard PCA. Across multiple equity portfolios sorted on size, book-to-market, accruals, investment, profitability, and other characteristics, heteroPCA achieves higher Sharpe ratios and lower average pricing errors. These improvements are economically meaningful, with Sharpe ratio gains often exceeding 50% relative to standard PCA. While heteroPCA exhibits slightly worse time-series fit, an expected consequence of filtering out idiosyncratic variance, the tradeoff appears favorable given the substantial improvements in risk-adjusted returns.

Second, heteroPCA identifies factors with clear economic interpretation. In cross-sections with well-known factor structures, such as portfolios sorted on size and accruals, heteroPCA shares the market factor with standard PCA and isolates a clean accrual factor that standard PCA leaves in a noisy third component. The improved interpretability stems from heteroPCA's ability to uncover systematic covariance patterns that remain hidden beneath idiosyncratic noise in standard PCA. By downweighting high-variance assets when their variance reflects idiosyncratic risk, heteroPCA focuses on true systematic co-movement.

Third, the stochastic discount factors implied by heteroPCA factors exhibit better pricing accuracy and more intuitive economic content. HeteroPCA consistently delivers lower Hansen-Jagannathan distances, and the SDF-mimicking portfolios display coherent patterns aligned with known risk premia. For instance, in size-accrual portfolios, heteroPCA identifies a clear quality-versus-junk trade, whereas standard PCA produces less

interpretable "size barbell" patterns.

These findings hold across a broad set of cross-sections, including double- and triple-sorts, asset pricing tree portfolios, anomaly portfolios, theme-aggregated factor portfolios, and, importantly, G10 currency portfolios, indicating that the value of correcting for heteroskedasticity is not specific to U.S. equities or to any single sorting scheme. Examining the mechanism, I find that heteroPCA leaves the dominant market factor essentially unchanged but rotates the weaker factors away from idiosyncratic-noise directions. The resulting loadings are more stable than PCA's in normal periods, with occasional large rotations only when the cross-sectional variance structure shifts abruptly. The higher-order heteroPCA factors reorient toward systematic dimensions that standard PCA leaves buried in idiosyncratic noise, most clearly the accrual dimension in the size-and-accruals portfolios. Throughout, the correction is achieved with nothing more than the return panel itself, and no characteristics, instruments, or auxiliary data are required.

These findings have practical implications for empirical asset pricing. Heteroskedastic idiosyncratic variances are pervasive across asset classes. In equities, firm size, leverage, and growth opportunities generate varying levels of idiosyncratic risk. In foreign exchange markets, interest rate differentials and liquidity differences create heterogeneous volatility across currencies, and similar patterns arise in bonds, commodities, and other markets. Even when assets are grouped into portfolios, as is standard practice, characteristic-driven volatility persists. These results show that correcting for this heteroskedasticity leads to more reliable and interpretable risk factors.

The heteroPCA methodology can be integrated into existing PCA-based approaches in asset pricing, including instrumented PCA, projected PCA, and risk premium PCA. By addressing heteroskedasticity as a first step, these methods may achieve improved factor estimation and better pricing performance. This paper deliberately restricts attention to the comparison with standard PCA, in order to isolate the contribution of the covariance correction on its own.

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Tables and figures

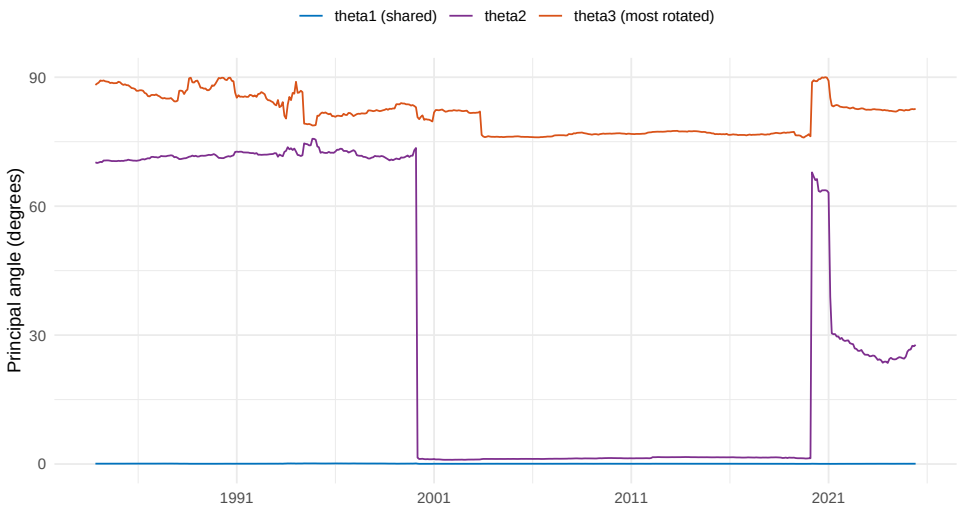


Figure 1: Principal angles between the PCA and heteroPCA factor subspaces over rolling windows (Size & Accruals). The smallest angle θ_1 stays near zero, so the two methods share one factor direction, the market. The two larger angles θ_2 and θ_3 are sizable, so heteroPCA reorients the two weaker factor directions away from PCA's. Because they describe the subspaces rather than individually labeled factors, the angles are unaffected by factor reordering.

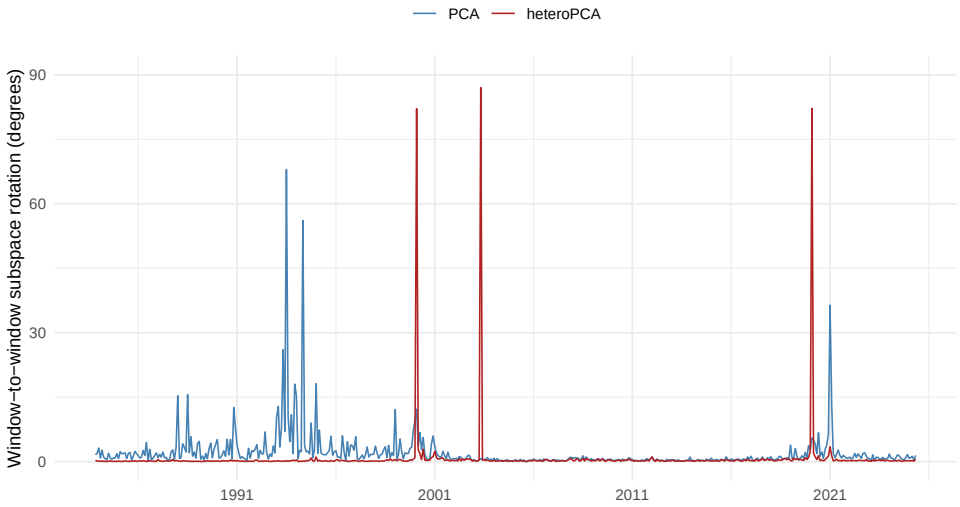


Figure 2: Window-to-window rotation of the estimated factor subspace (Size & Accruals). For each method the line is the largest principal angle between the $K = 3$ factor subspace estimated in a rolling window and the one estimated in the previous window. HeteroPCA is more stable than PCA in normal times and rotates sharply only at a few structural breaks. Because the measure compares the subspaces rather than individually labeled factors, it is unaffected by factor reordering.

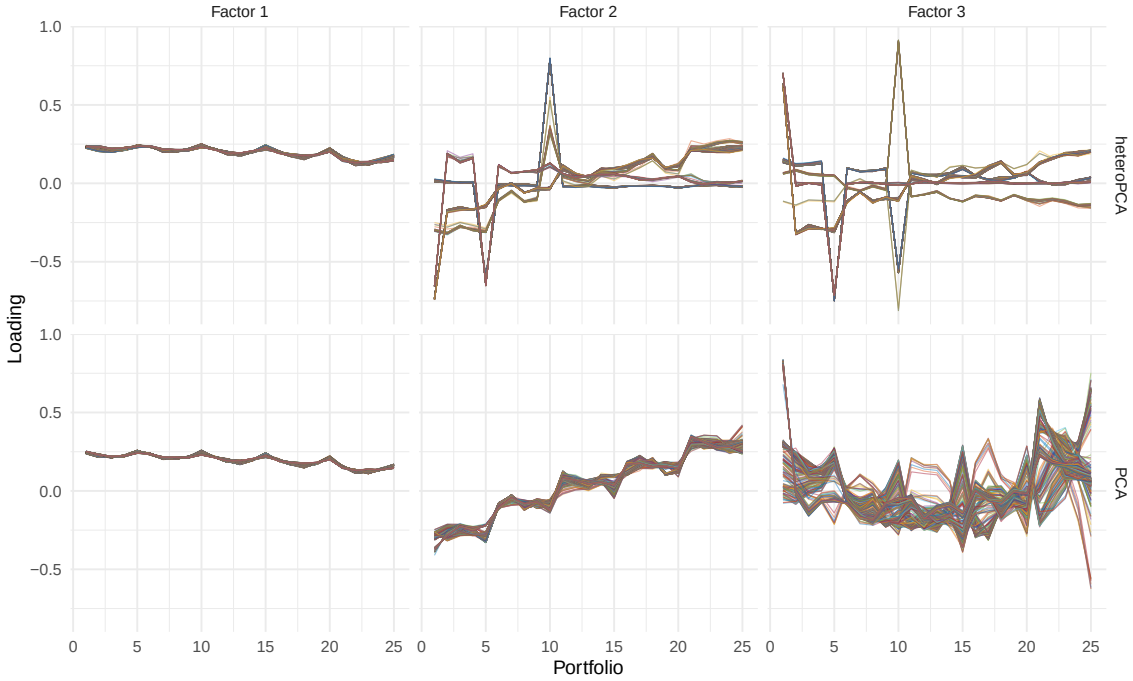


Figure 3: Time-stability of factor loadings (Size & Accruals). For each factor, every line is the loading vector estimated on one rolling window, across the portfolios, with heteroPCA in the top row and PCA in the bottom row. The rotation is calculated at each window to minimize the distance between the window's loadings and those estimated over the whole sample, so that tightly overlapping lines indicate loadings that are stable through time.

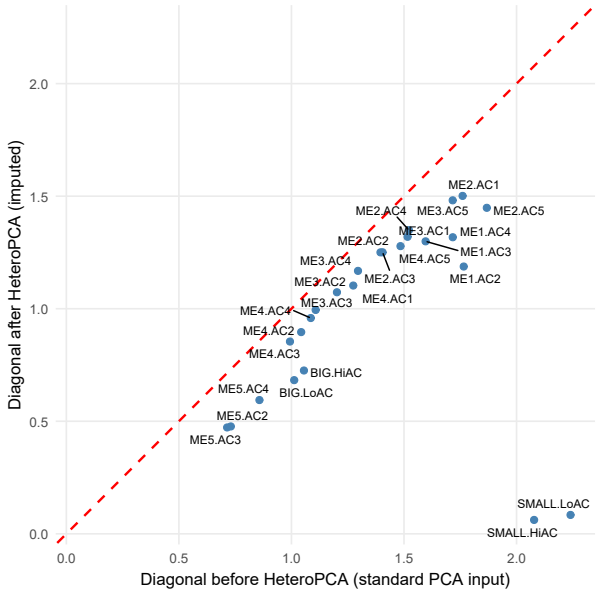


Figure 4: Diagonal adjustment for Size & Accruals portfolios, full sample. The horizontal axis is the original diagonal (total variance) and the vertical axis is the covariance-implied diagonal from heteroPCA. HeteroPCA flattens the diagonal, especially for the small, extreme-accrual portfolios.

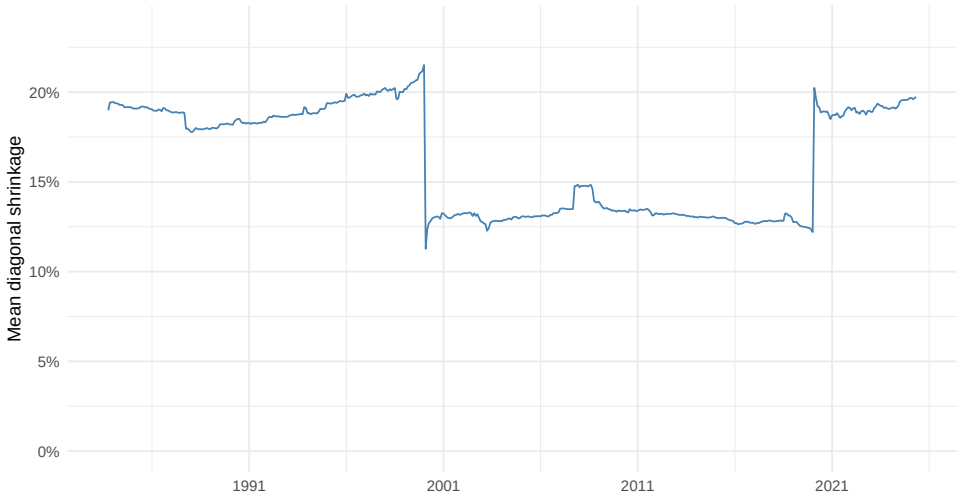


Figure 5: Mean diagonal shrinkage applied by heteroPCA over time (Size & Accruals). The line shows the average fractional reduction of the covariance diagonal across rolling windows.

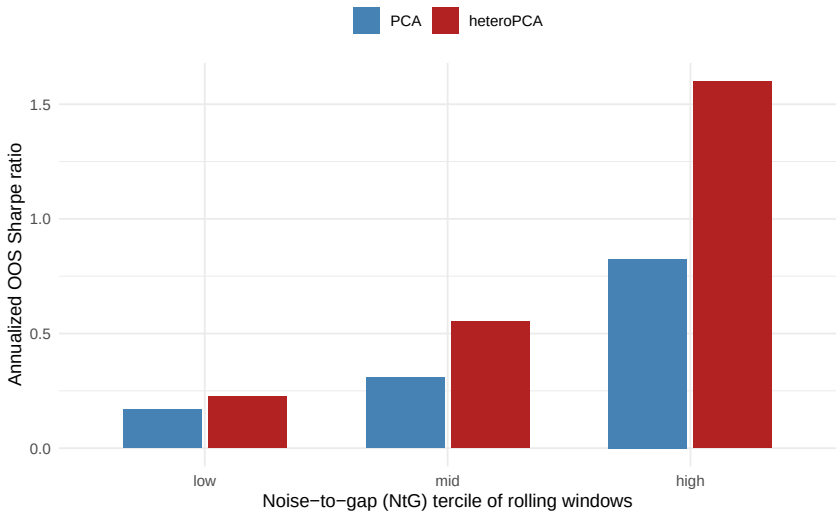


Figure 6: Out-of-sample Sharpe ratio by noise-to-gap regime (Size & Accruals). The rolling windows are split into terciles of the noise-to-gap ratio, defined as the average covariance diagonal relative to the eigengap between the third and fourth eigenvalues. Within each tercile the bars report the annualized out-of-sample Sharpe ratio of the three-factor mimicking portfolio under PCA and heteroPCA. The heteroPCA advantage widens sharply in the high-ratio windows, where the leading eigenvectors are least well identified.

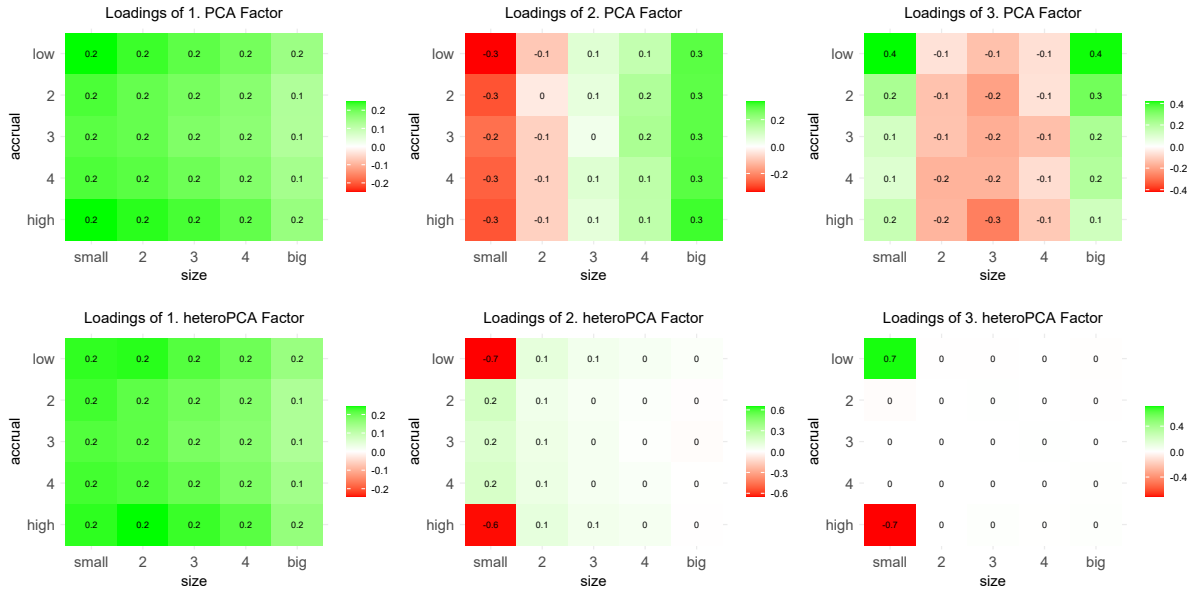


Figure 7: Factor loadings for Size & Accruals portfolios. Top panel: PCA (top 3 factors). Bottom panel: heteroPCA (bottom 3 factors).

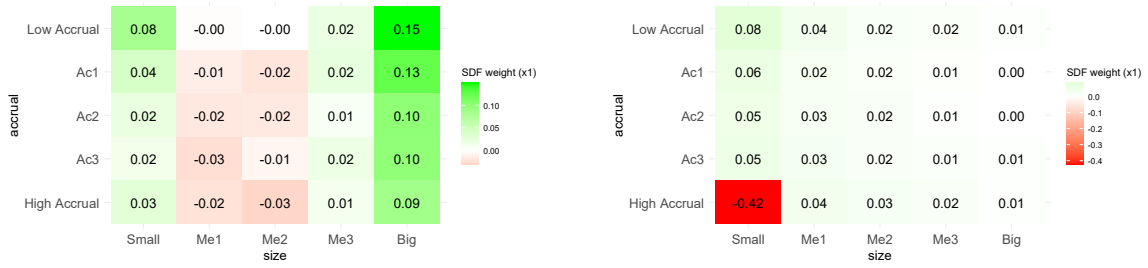


Figure 8: SDF-mimicking portfolio weights for Size & Accruals portfolios. Left panel: PCA exhibits a "size barbell" pattern. Right panel: heteroPCA shows a clear "quality versus junk" pattern.

Table 1: Out-of-sample fit: $K = 3$ factors. T_0 set to 5 in heteroPCA case.

	SR		RMS_α		$\bar{\sigma}_e^2$ (%)	
	heteroPCA	PCA	heteroPCA	PCA	heteroPCA	PCA
SIZE&BM	**0.28**	0.15	**0.12**	0.16	9.09	**7.58**
SIZE&ACC	**0.22**	0.13	**0.09**	0.10	7.25	**6.83**
SIZE&INV	**0.31**	0.20	**0.12**	0.14	8.62	**7.25**
SIZE&OP	**0.20**	0.11	0.14	**0.12**	10.17	**8.70**
SIZE&STREV	0.03	**0.12**	0.18	0.18	10.18	**7.65**
SIZE&MOM	**0.28**	0.18	**0.13**	0.17	9.39	**8.04**
SIZE&IVOL	**0.29**	0.21	**0.15**	0.17	7.85	**6.09**
SIZE&VOL	**0.29**	0.22	**0.14**	0.15	7.66	**6.03**
OP&INV	**0.24**	0.21	**0.12**	0.13	18.83	**17.37**
SIZE&BM&INV	**0.20**	0.11	**0.16**	0.18	12.81	**12.40**
SIZE&BM&OP	**0.24**	0.16	**0.14**	0.20	20.47	**18.03**
SIZE&OP&INV	**0.32**	0.18	**0.12**	0.18	13.32	**12.67**
AP-Tree (Tree10)	**0.52**	0.18	**0.68**	0.85	9.77	**9.62**
AP-Tree (Tree40)	**0.57**	0.26	**0.76**	0.90	11.25	**10.48**
KNS (extreme deciles)	**0.18**	0.12	**0.22**	0.23	15.86	**15.81**
JKP (themes)	**0.21**	0.10	0.07	0.07	1.28	**1.11**
FX (G10 extremes)	**0.12**	0.04	0.01	0.01	16.97	**15.96**
Individual stocks	0.14	**0.15**	**0.38**	0.39	72.17	**71.96**

Table 2: Out-of-sample Sharpe-ratio gain of heteroPCA over PCA by noise-to-gap regime. For each cross-section the rolling windows are split into terciles of the noise-to-gap ratio. Each entry is the difference in the annualized out-of-sample Sharpe ratio of the three-factor mimicking portfolio, heteroPCA minus PCA, within that tercile. The column "NtG top" reports the median noise-to-gap ratio in the high tercile, a measure of how high the ratio reaches in that cross-section. Cross-sections are ordered by this reach. The panels are those for which the rolling noise-to-gap series is available.

Cross-section	NtG top	Low	Mid	High
Size & Accruals	53.4	0.06	0.25	0.78
JKP themes	14.8	0.42	0.30	0.53
Size & Profitability	9.8	-0.18	0.04	1.02
Size & Investment	6.7	0.00	0.40	0.54
AP-Tree (10)	5.7	0.01	0.99	2.62
Size & OP & Inv	5.6	0.28	0.06	0.95
AP-Tree (40)	3.9	0.59	0.66	2.03
Size & Volatility	3.9	-0.03	-0.03	0.63
Size & BM	3.3	0.73	0.32	0.49
Size & IdioVol	2.9	-0.08	0.40	0.42
Size & Momentum	2.5	0.84	0.02	0.44
KNS (extreme)	2.4	-0.01	0.05	0.57
Size & ST-Reversal	2.3	0.02	-0.53	-0.40
Size & BM & Inv	2.3	0.36	0.53	-0.18
FX (G10)	1.9	0.45	0.06	0.23
Size & BM & OP	1.9	0.02	0.27	0.76
Median		0.04	0.26	0.55

Table 3: Hansen-Jagannathan distance. $K = 3$ factors. T_0 set to 5 in heteroPCA case.

	heteroPCA	PCA
SIZE&BM	**0.27**	0.35
SIZE&ACC	**0.26**	0.31
SIZE&INV	0.41	**0.37**
SIZE&OP	0.24	**0.23**
SIZE&STREV	0.33	**0.31**
SIZE&MOM	**0.29**	0.35
SIZE&IVOL	**0.34**	0.38
SIZE&VOL	**0.33**	0.38
OP&INV	0.24	0.24
SIZE&BM&INV	**0.25**	0.34
SIZE&BM&OP	**0.23**	0.30
SIZE&OP&INV	**0.30**	0.36
AP-Tree (Tree10)	**1.42**	1.47
AP-Tree (Tree40)	**1.42**	1.47
KNS (extreme deciles)	0.74	0.74
JKP (themes)	**0.72**	0.73
FX (G10 extremes)	**0.22**	0.24
Individual stocks	0.57	0.57

Appendix

A. Deep dive: the FX currency cross-section

This appendix repeats the factor-level comparison of Subsection 4.2 for the G10 currency portfolios, the cross-section furthest from the US equity panels that motivate the method. The aim is to show that the patterns documented for Size & Accruals also hold for an asset class with a very different return structure.

Table A1 reports the full-sample principal angles between the PCA and heteroPCA factor subspaces for the currency cross-section. As in the equity case the two methods agree on the dominant directions, with the two smallest principal angles near zero, and differ on a single direction, the largest principal angle being about seventy-nine degrees. The comovement of currency baskets is very low-dimensional, so heteroPCA reorients only the weakest factor direction.

Table A1: Principal angles (degrees) between the PCA and heteroPCA factor subspaces, full sample (G10 currency cross-section).

	θ_1	θ_2	θ_3
FX (G10 extremes)	0.4	2.8	78.6

Figure A1 and Figure A2 show the full-sample factor loadings and the SDF-mimicking portfolio weights for the currency baskets. The reorientation of the third factor is economically sharp. Under heteroPCA the third factor and the implied discount factor load almost entirely on the high-carry basket, whereas PCA spreads weight thinly across many baskets and even takes the opposite sign on the carry dimension. HeteroPCA concentrates the discount factor on the systematic carry dimension rather than on noisy baskets.

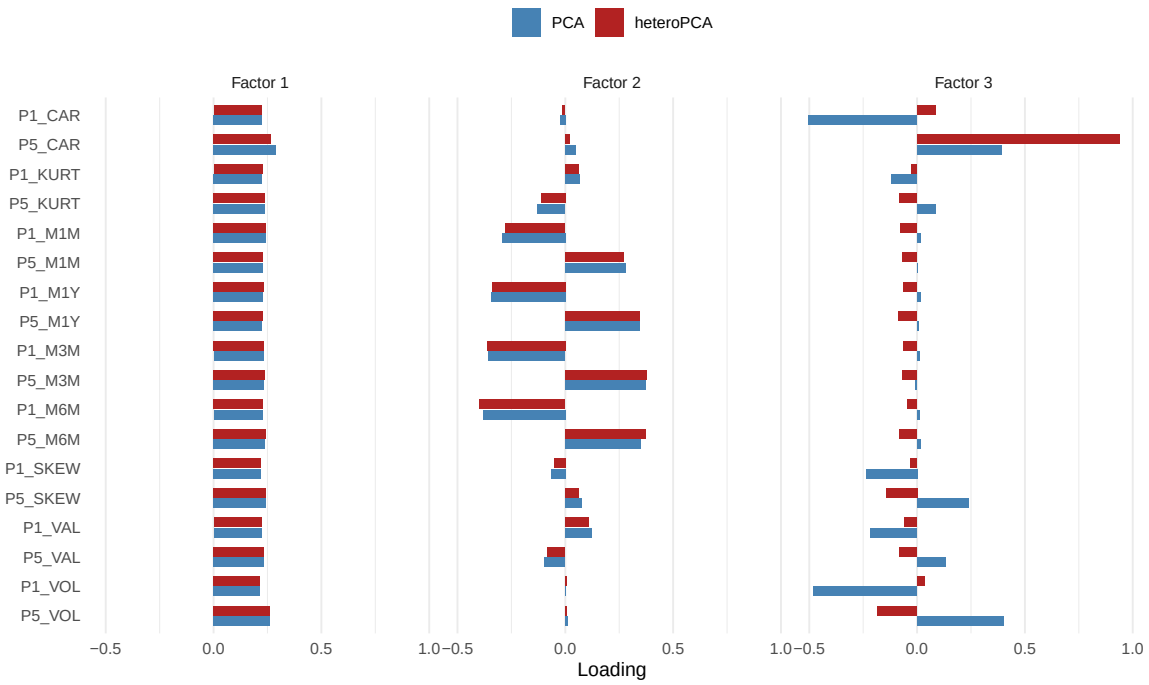


Figure A1: Factor loadings for the G10 FX extreme portfolios, PCA versus heteroPCA, full sample.

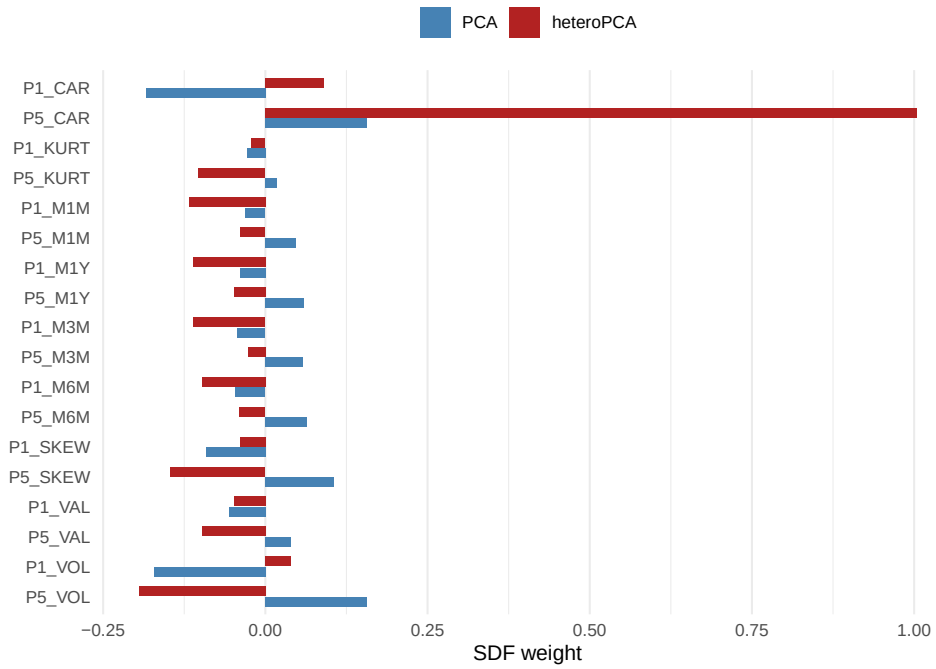


Figure A2: SDF-mimicking portfolio weights for the G10 FX extreme portfolios, PCA versus heteroPCA, full sample.